

Security and Privacy Concerns in Federated Learning: An Introduction

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Agenda

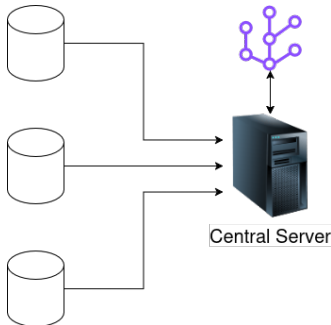
- Federated Learning
 - Motivation
 - Training Process
 - Limitations
- Decentralized Federated Learning
 - Blockchain
- Threats
 - Security
 - Privacy

Federated Learning: Motivation

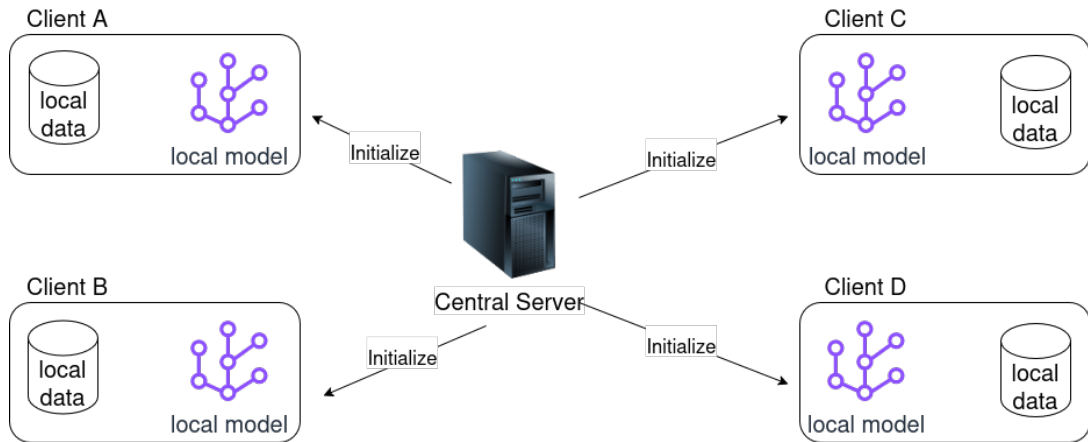
- Training one central model on distributed data
- Transmitting full datasets causes high communication cost
- Data may contain sensitive information (medical data, PII)

Solution

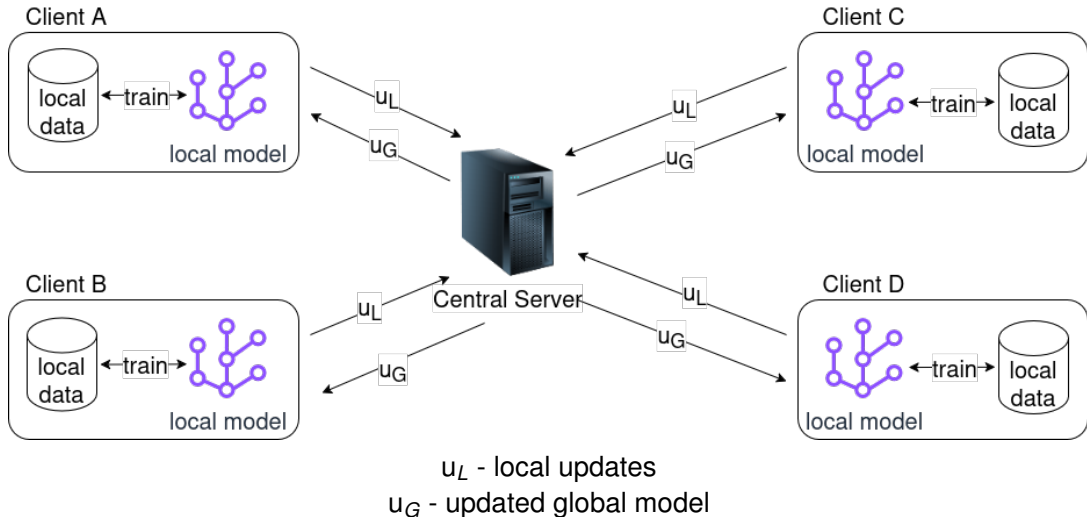
Train locally on datasets and only communicate updates



Federated Learning: Initialization of local models



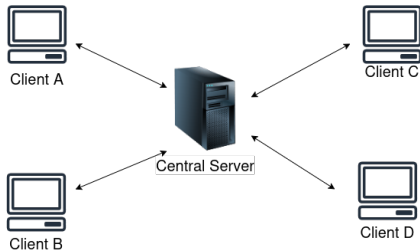
Federated Learning: Training



Federated Learning

1. Initialize global model
2. Distribute model to clients
3. Clients perform local training
4. Server aggregates updates
5. Share updated model

Repeat 2.-5. until convergence

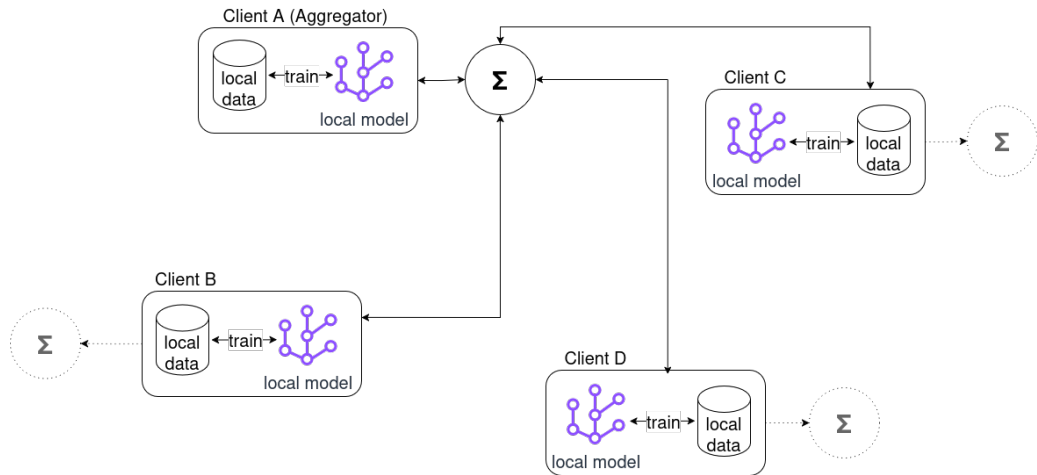


Main Limitation: Dependence on Server

Requires trust in the central server

Scalability bottleneck

Decentralized Federated Learning

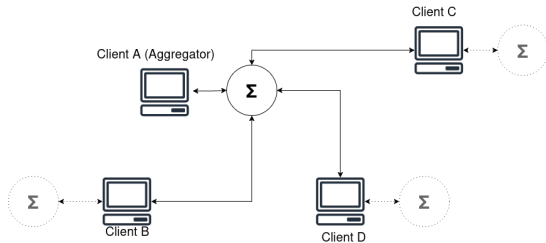


Decentralized Federated Learning

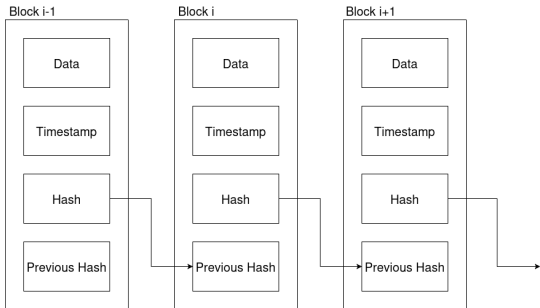
1. Local training and model aggregation as in centralized FL
2. Clients take turns acting as aggregators
3. Model updates are exchanged via peer-to-peer communication
4. Commonly uses blockchain to ensure traceability and immutability

Main Risks

Blockchain-related vulnerabilities
Malicious clients



Blockchain

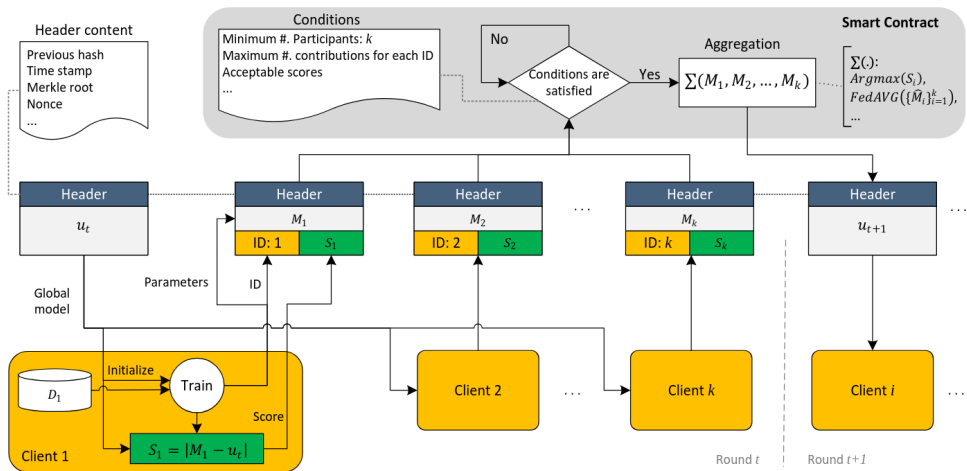


- Tamper-resistant, append-only log storing model updates and aggregation steps
- Decentralized coordination using Smart contracts to enforce participation rules and aggregation conditions

Limitations

Latency and computational overhead
Continuous storage growth
Known vulnerabilities

Blockchain-based DFL [1]



Blockchain-specific Vulnerabilities

- Communication disruption (DDoS)

Fill up the blockchain buffer with spam data

- Consensus Attacks

Create multiple fake nodes to influence control consensus mechanism

- Routing Attacks

Manipulate package routing to change blockchain structure

- Privacy poisoning

Insert personal data into a blockchain block, causing non-compliance with legal requirements

Attacks on Model Performance

- Data Poisoning

Train local model on artificially created dataset to inject backdoors, then share poisoned gradients

- Model Poisoning

Maliciously influence global model training, e.g. by changing the objective function

- (Evasion Attacks)

Decrease model performance during inference by manipulating input data. Can be facilitated by poisoning attacks during training, but is not inherently tied to FL

- **Model Inversion**

Approximate client data using generative models, e.g. GANs

- **Inference Attacks:**

Infer information about a dataset based on the model updates generated by a client using that dataset

- Feature Inference: Gain information about sensitive features
- Membership Inference: Determine if a data point was used to train the model

References

- [1] Ehsan Hallaji u. a. “Decentralized Federated Learning: A Survey on Security and Privacy”. In: *IEEE Transactions on Big Data* 10.2 (Apr. 2024), S. 194–213. ISSN: 2372-2096. DOI: 10.1109/tbdata.2024.3362191. URL: <http://dx.doi.org/10.1109/TBDATA.2024.3362191>.